



LULING – TEXAS
Enhanced Oil Recovery (EOR) Program Report
RDV® – Dynamic Vasoactive Reactor

Context

Mature light crude oil field in East Texas. Stripper wells with average production below 0.5 BPD, operating at negative margins and approaching abandonment. The identified problem was organic deposition (paraffins) restricting permeability. Conventional solvent treatments provided temporary relief but failed to address the root cause.

Design

A matrix stimulation intervention was designed for EOR purposes using RDV®. A pill was prepared at the surface consisting of a 50% RDV® and 50% crude oil blend extracted from the same well. Once the pill was placed above the wellbore fluid level, a 24-hour recirculation cycle was planned, followed by a soaking period to allow product permeation into the formation, leveraging the low differential pressure drive from the reservoir toward the wellbore.

Special attention was given to the integrity of well completions in certain wells, whose structural weakness precluded forced injection (squeeze) of the treatment into the formation. The objective was to facilitate the activation of the protonation reaction, which induces molecular fragmentation of hydrocarbons present in the wellbore fluid column and mitigates organic deposits obstructing formation pore throats.

Application

A group of seven stripper wells were treated in the Luling field. In the most extensively documented case (Well Luling #P1), the well was producing 0.25 BPD prior to treatment. The RDV® + crude oil pill was placed above the wellbore fluid level; however, the beam pump failed before the recirculation cycle stipulated in the protocol could be executed. Consequently, the well remained shut-in for two years, with the pill in a static soak but exposed to the bottomhole pressure and temperature conditions required to activate the chemical protonation reaction. Following this period, workover operations were performed to repair the pump. Once repaired, and without any further recirculation, the well was placed back on production, yielding 42 BPD.

Results

Parameter	Value
Wells treated	7
Average production before	<0.5 BPD
Average increase	>500%
Best documented case	0.25 → 42 BPD (>8,400%)
Duration of effect	Months (prolonged residual effect)
Estimated ROI	2–14 days (Luling #P1 case)

Conclusions

The molecular fragmentation induced by RDV®, through the irreversible cleavage of C–C and C–H bonds in paraffin chains, restored the effective porosity of the formation and eliminated the organic constriction restricting hydrocarbon flow toward the wellbore. Consequently, relative permeability to oil improved significantly, re-establishing the productive capacity of the rock–fluid system.

In Well Luling #P1, the results demonstrated that RDV® (Dynamic Vasoactive Reactor) can sustain its chemical effect over extended periods of operational downtime. The protonation reaction remained active throughout the two-year shut-in period, relying exclusively on the natural reservoir energy (bottomhole pressure and temperature) as a catalyst, without requiring recirculation or product re-application.

Technical Department

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